
What Do Self-Supervised Pose Representations Add Beyond Anatomy? Lessons for Body Foundation Models from Gait Laterality

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Abstract

1 Foundation models for the brain and body need features that preserve movement
2 information relevant to motor function. We develop a controlled test using gait
3 laterality: a measure derived from left–right differences in landmark speed that
4 reverses sign when the sides are exchanged. We use 625 gait clips from 93
5 YouTube videos drawn from the Gait Abnormality in Video Dataset (GAVD), a
6 manually curated resource for clinical gait analysis. Our quality-screened subset
7 includes annotations for normal gait and several neurological and muscular con-
8 ditions. We compare a compact pose encoder, trained by self-supervised latent
9 prediction, with its matched untrained version while holding the body-landmark
10 selection and prediction method fixed. Test videos remain separate, and training is
11 repeated from different starting weights. Mirrored training examples improve fea-
12 ture consistency under left–right exchange relative to training without them, but
13 our main prediction tests show no clear gain from either pretraining or mirroring.
14 A fixed prediction rule guarantees the expected sign change even with features
15 from an untrained encoder; with this rule held fixed, pretraining without mirrored
16 examples produces poorer predictions. The study contributes an evaluation that
17 distinguishes anatomically consistent behavior from evidence that self-supervised
18 pretraining improves movement prediction. These experiments on a compact en-
19 coder motivate testing whether larger body models gain predictive value beyond
20 their built-in anatomical structure. Connecting such comparisons to independently
21 measured gait is necessary before assigning physiological meaning to the learned
22 features.

23 1. Introduction

24 A long-term goal of foundation models for the brain and body is to learn representations that retain
25 meaningful aspects of motor function. For gait, this would mean that the model’s *latent space*—its
26 internal numerical description of movement—preserves useful relationships between body parts and
27 over time, while remaining reliable across recording conditions. Clinical studies provide concrete
28 examples of what such representations might preserve: step length and the duration of stance and
29 swing are used to describe post-stroke gait asymmetry, while the timing of successive left and right
30 steps provides a distinct measure of coordination [10, 9].

31 Video-estimated body landmarks make these questions accessible without laboratory motion cap-
32 ture. For the laterality evaluation, the encoder receives 33 landmarks and predicts hidden features
33 at 12 specified locations: the shoulders, hips, knees, ankles, heels, and foot tips. The shoulders
34 and hips provide references for body alignment, while the remaining locations describe lower-limb

35 movement. Clinical gait research motivates investigating this anatomical selection, but we have not
36 established that it is optimal or that its features measure neurological impairment.

37 We use *laterality*, defined here as a signed difference in estimated movement between the left and
38 right sides, to test what the features preserve. The target averages speed contrasts from five landmark
39 pairs. Exchanging left and right must reverse its sign, giving us a known response against which to
40 test the model. We can evaluate both prediction of the movement quantity and the features’ response
41 to reflection, without assuming that either measures clinical impairment.

42 The central question is what learned pose features add beyond the body structure and movement
43 information already available to the model. We compare a compact, pretrained S-JEPA-inspired en-
44 coder with its own untrained starting weights, holding the architecture and anatomical feature con-
45 struction fixed [1]. Mirrored training examples and a sign-constrained prediction rule let us examine
46 how anatomical knowledge affects the features and their predictions. An exploratory condition-label
47 classification check compares learned features with simple pose summaries. Together, these com-
48 parisons connect body-aware model design to evidence of predictive value, using baselines suited to
49 each task.

50 2. Background and related work

51 Joint-embedding predictive architectures learn by predicting hidden feature representations. I-JEPA
52 applies this approach to images, and S-JEPA applies it to skeleton sequences [2, 1]. Our laterality
53 encoder changes the architecture and training objective, so its results concern the tested implementa-
54 tion and cohort. Related gait work, including GaitForeMer, evaluates self-supervised representations
55 against gait-impairment severity [5]. The coordinate-derived movement quantity used here offers a
56 controlled test before assigning clinical meaning to the representation.

57 Anatomical structure can enter a model through its layers, training examples, or output rule. Group-
58 equivariant networks and frame averaging provide established ways to control responses to transfor-
59 mations [4, 11]. Reflection augmentation encourages consistency through training, while the odd
60 readout used here guarantees a sign change algebraically. Clinical symmetry research motivates
61 examining the two sides separately [10], but does not validate our particular speed contrast. The
62 experimental question is what these choices reveal about information accessible from the learned
63 features.

64 3. Methods

65 3.1 Data and evaluation on unseen videos

66 The Gait Abnormality in Video Dataset (GAVD) is a manually curated resource for clinical gait
67 analysis. Its creators sourced relevant online videos, and clinicians screened and annotated the
68 walking sequences [12]. We use pose extracted from a subset of its YouTube videos with normal,
69 Parkinson’s, stroke, myopathic, or cerebral-palsy annotations. A *sequence* is a short annotated clip; a
70 *source* is the original uploaded video. Quality and target-computability checks retain 625 sequences
71 from 93 sources out of 642 sequences with extracted pose. We form this eligible cohort before
72 assigning source folds. We take the dataset’s condition categories as given and make no independent
73 clinical assessment of them.

74 We divide the 93 videos into five groups, approximately balancing annotations. Each round trains on
75 four groups and tests on the remaining group, with every clip from a video kept together. These are
76 the five *outer folds*: each video is test data once per seed. Five *seeds* vary the random initialization
77 and training draws, producing 25 trained encoders per training variant and 50 in total. Grouping
78 reduces shared-recording dependence [13]; reliable person identities are unavailable, so the same
79 individual may still appear in different groups.

80 An exploratory condition-label classification check compares learned features with simple pose sum-
81 maries on 20 test videos. It uses a separate encoder trained by introducing annotation categories in
82 stages, with one split and one initialization. Appendix B describes its methods and overlapping
83 cohort.

84 3.2 Pretrain the laterality encoder

85 Each clip is represented by 33 pelvis-centered, scale-normalized landmarks over 64 frames. Four
86 consecutive XYZ observations at one landmark form a patch, and a *token* is the encoder’s represen-
87 tation of that patch. During training, up to 60% of the eligible patches at the 12 specified landmarks
88 are hidden. Appendix A.1 gives the validity rules that decide which patches are eligible.

89 A four-layer, 96-dimensional encoder and a two-layer Transformer predictor estimate the hidden
90 features provided by a slowly updated target encoder. The loss combines centered latent cross-
91 entropy with a two-view VICReg-style regularizer [3]. Neither the laterality target nor the annotation
92 label enters this pretraining loss.

93 The variant without mirrored training clips never reflects a sample, while the variant with mirrored
94 training clips reflects each sample with probability 0.5 before generating two training views. The
95 two variants share their initialization, source-draw schedules, and update counts, so the reflection
96 augmentation is the only intended difference between them. Both receive 1,200 optimizer updates
97 without early stopping. Appendices A and C give the training details and pipeline overview.

98 3.3 Define the left–right movement quantity

99 Reflection M negates the horizontal coordinate and exchanges anatomical left/right landmark in-
100 dices, including validity flags. Applying it twice restores the input, so $M^2 = I$. For the shoulders,
101 knees, ankles, heels, and foot tips, we measure median left and right speeds $m_{L,k}$ and $m_{R,k}$ on the
102 same valid frame transitions:

$$c_k = \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + 10^{-8}}, \quad y(X) = \frac{1}{5} \sum_{k=1}^5 c_k.$$

103 A positive pair contrast means the left landmark has the higher median speed on that support. Every
104 pair requires at least eight transitions with both sides observed at both endpoints. Speeds use orig-
105 inal timestamps and uninterpolated normalized coordinates. Hips define pelvis centering and are
106 excluded from this target because their two centered speed magnitudes coincide.

107 The construction gives $y(MX) = -y(X)$ for any eligible clip, including one with asymmetric
108 movement. The target has mean -0.006 and standard deviation 0.059. It is an estimated-coordinate
109 motion index; its relation to clinical gait measurements remains untested.

110 3.4 Compare predictions with and without enforced sign reversal

111 A *readout* predicts laterality from encoder features using fitted regression. Left/right mean tokens
112 from five landmark pairs, pooled over shared valid times, yield five differences and five sums. Con-
113 catenating these gives a 960-value vector $A(X)$ for pretrained and untrained encoders.

114 The paper reports two readouts. The first predicts directly from $A(X)$, with no constraint on how
115 the prediction responds to reflection; this is the baseline against which pretraining is judged. The
116 second builds the sign reversal into the readout itself. Splitting $A(X)$ into the part that flips sign
117 under reflection and the part that does not,

$$\Phi^-(X) = \frac{A(X) - A(MX)}{\sqrt{2}}, \quad \Phi^+(X) = \frac{A(X) + A(MX)}{\sqrt{2}},$$

118 the odd part Φ^- reverses sign under reflection while the even part Φ^+ stays unchanged. A readout on
119 the odd part with feature centering and the intercept both disabled, $h(X) = w^\top \Phi^-(X)$, guarantees
120 $h(MX) = -h(X)$ for any encoder and any weights, whatever its accuracy. This is what lets us ask
121 whether correct sign behavior, supplied by construction, carries any predictive benefit on its own.

122 Encoder weights stay fixed while separate regressions are fitted to pretrained and untrained features.
123 Four inner folds of outer-training videos select each regression penalty; they tune only the readout,
124 although the encoder can already have seen these inputs during pretraining. Appendix A.3 records
125 two further readouts we examined but do not report in the main comparison.

3.5 Measure feature agreement and uncertainty

We compare the encoder’s features for a clip and its left–right-mirrored copy after matching body landmarks—for example, the mirrored left ankle with the original right ankle. The disagreement score q compares these values directly, without fitted alignment. Zero means complete agreement; larger scores mean more disagreement. A feature that barely changes across clips will also agree closely with its mirror, so a low score is consistent with either genuine symmetry or near-constant features, and prediction accuracy is needed to tell them apart. Appendix A gives the equation and validity checks.

For prediction, clips receive weights inverse to their source’s clip count, so each video has equal total weight. For each seed, we pool test predictions from all five folds and calculate weighted sequence-level R^2 , then average the five scores. An R^2 of zero matches the weighted-mean-target reference for the scored data; negative values indicate larger squared error. This differs from averaging targets within a video before scoring, or averaging predictions across seeds.

The 2,000 bootstrap samples redraw the 93 source videos with replacement, carrying all their clips and seeds together. Both conditions in each paired comparison use the same draws. The resulting 95% intervals are pointwise and conditional on the fitted models and fixed split. They omit uncertainty from retraining, alternative splits, and choices made during development. Laterality estimates and intervals use three decimal places to retain small differences; calculations and relative reductions use unrounded values.

4. Results

4.1 Mirrored training improves feature consistency

Mirrored examples reduce feature disagreement by about 7% compared with training without them (reduction 0.008; 95% interval [0.007, 0.010]). The video-balanced score is lower for each of the five initializations. The features therefore agree more closely between original and mirrored clips after matching left/right landmarks. However, both pretrained variants still have greater average disagreement than their matched untrained versions; the improvement is between the two training variants.

4.2 Prediction from learned and baseline features

Without enforced sign reversal, pretraining without mirrored examples yields $R^2 = 0.060$ (95% interval [-0.025, 0.126]) on unseen videos. Table 1 separates the effects of pretraining and mirrored examples, including a comparison in which the sign-reversal rule is held fixed.

Table 1: Differences in laterality prediction; positive values indicate improvement. Rows 1 and 3 compare the encoder pretrained without mirrored clips with its matched untrained version; row 2 compares training with versus without mirrored clips. Rows 1 and 2 do not enforce sign reversal. Intervals are pointwise 95% source-bootstrap intervals conditional on fitted models (Appendix A.4).

Question	Difference in R^2 (95% interval)	What the result supports
Does pretraining improve prediction?	-0.018 [-0.039, 0.002]	No clear improvement
Do mirrored training examples improve prediction?	+0.004 [-0.006, 0.013]	Improvement remains uncertain
With the sign rule fixed, does pretraining improve prediction?	-0.059 [-0.095, -0.017]	Pretrained encoder scores lower

The first two intervals include zero: neither pretraining nor mirroring shows a clear prediction gain. The sign-reversal rule in the third comparison holds for pretrained and untrained encoders, with numerical checks passing for every seed. Under this rule, the encoder pretrained without mirrored

160 clips scores 0.059 lower in R^2 than its matched untrained version; the entire interval is below zero.
161 Correct sign behavior alone therefore cannot establish useful pretraining.

162 Simple input summaries provide another baseline for assessing learned features. In the exploratory
163 condition-label classification check, raw pose summaries correctly classify 10 of 20 test videos, com-
164 pared with 6 of 20 using learned features. This result supports retaining an input-based comparison
165 for that task, although the single-split design and absence of a matched untrained encoder prevent
166 attributing the difference specifically to pretraining. Appendix B reports balanced accuracy and a
167 landmark-availability comparison.

168 5. Implications for physiologically meaningful representations

169 The laterality comparisons isolate what pretraining adds, with the anatomical inputs, feature con-
170 struction, and form of the prediction rule all held constant. On that footing we can compare features
171 from pretrained and matched untrained encoders, including the case where correct left-right sign
172 behavior is supplied by design rather than learned. Mirrored examples improve feature agreement
173 under reflection without a clear prediction gain. The classification check asks whether learned fea-
174 tures beat simple summaries of the input; its result argues for keeping a pose baseline, though it
175 leaves open why the learned features do worse.

176 A physiologically meaningful latent space remains a research objective. Progress toward it would
177 require testing whether features preserve independently measured quantities such as step timing,
178 joint motion, or bilateral coordination across people and recording conditions. Our signed speed
179 contrast offers a controlled intermediate test, but cannot substitute for those measurements. Adding
180 a laterality-supervised loss would also be a new experiment: the laterality encoder receives anatomy
181 through its inputs, target selection, and augmentation, without a clinical or laterality label in its loss.

182 6. Strengths, limitations, and responsible use

183 Our laterality evaluation builds on GAVD’s curated recordings with matched training comparisons
184 [12]. Several safeguards support it: test videos and all their clips are excluded from fitting and tuning,
185 which addresses dependence between clips from the same recording [13]; repeated training across
186 paired initializations reduces reliance on any single starting point; and each video receives equal
187 scoring weight, with the bootstrap resampling whole videos to retain within-recording dependence
188 (Section 3, Appendix A.4).

189 These safeguards support controlled laterality comparisons on unseen source videos within GAVD,
190 but they do not add independent recordings. Person identities are unavailable, so performance on
191 reliably separated participants and an external cohort remains to be established. Estimated pose
192 and the speed index also require validation against measured gait before clinical interpretation. The
193 findings concern the tested encoder, its linear prediction rules, and the specified reflection test, and
194 the classification encoder’s training settings have not been independently checked (Appendix B).

195 GAVD distributes its annotations under the MIT License, permitting reuse, including academic re-
196 search, subject to retaining the required copyright and permission notices [6]. The source videos are
197 hosted separately; GAVD’s access guidance specifies platform terms and ethics and data-protection
198 requirements for their use [7]. Our study-specific ethics, video-use, and derived-pose release re-
199 views remain unresolved, so submission and data release remain on hold. The experiments provide
200 no basis for diagnosis, surveillance, or clinical deployment.

201 7. Conclusion

202 Gait laterality offers a practical starting point for testing what movement information a body rep-
203 resentation preserves. Adding mirrored clips improved feature consistency without a clear gain
204 in predicting left-right movement differences. The expected sign change could also be built into
205 the prediction rule with features from an untrained encoder, so correct symmetry alone is no evi-
206 dence that training helped. For body foundation models, this makes predictive comparisons against
207 matched untrained features the test that judges what self-supervised training adds.

208 **Appendix A. Laterality methods and interpretation**

209 **A.1 Prepare the gait clips**

210 Quality checks and the requirement to calculate laterality retain 625 clips from 93 GAVD videos.
211 The annotations comprise 270 normal, 39 Parkinson’s, 75 stroke, 183 myopathic, and 58 cerebral-
212 palsy clips. We assign videos to the five evaluation groups after these checks, keeping every clip
213 from a video in the same group.

214 Coordinates follow the 33-landmark MediaPipe scheme [8]. An observed landmark must have finite
215 coordinates and a visibility score of at least 0.45. To calculate laterality, both hips must be observed
216 so coordinates can be centered at their midpoint; body size is normalized using the median three-
217 dimensional shoulder and hip widths. We use original timestamps and do not fill missing coordinates
218 for this target. Each of the five landmark pairs needs at least eight frame-to-frame intervals with both
219 sides observed at both ends, ensuring that left and right speeds use the same observations.

220 For encoder inputs, gaps of up to four missing frames between valid observations may be filled by
221 interpolation, and clips are resized to 64 frames. We track which observations remain usable and set
222 invalid coordinates to zero. A four-frame patch is used only when all four observations are marked
223 valid. A clip must provide at least 50% usable observations across the 12 target landmarks and
224 at least two complete target-landmark patches after this preparation; filled gaps can contribute to
225 coverage. The advantage of this landmark selection over alternatives has not been tested.

226 **A.2 Train with and without mirrored clips**

227 Both training conditions use a four-layer encoder with 96 feature values per patch and a two-layer
228 predictor, each with four attention heads. The predictor estimates hidden features supplied by a
229 slowly updated copy of the encoder. The training objective is

$$L_{\text{pose}} = L_{\text{latent CE}} + 0.05 (25L_{\text{invariance}} + 25L_{\text{variance}} + L_{\text{covariance}}).$$

230 The first term measures how well the predicted features match their targets. The remaining terms
231 compare two slightly perturbed views of the same clip: invariance encourages their features to
232 agree, variance discourages nearly constant features, and covariance discourages redundant feature
233 dimensions. The coefficient pattern 25, 25, and 1 follows VICReg [3]; the variance and covariance
234 penalties are averaged across the two views. The multiplier 0.05 sets their overall scale. These
235 weights are fixed for both conditions, and their suitability for gait has not been tested through a
236 sensitivity analysis.

237 Each encoder receives 1,200 AdamW updates with batches of 20 and an initial learning rate of 10^{-3} .
238 Each training epoch draws one clip per training video, then samples additional videos uniformly to
239 fill the batches. The two conditions share starting weights, video draws, and training length. Both
240 use small rotations and translations; one also mirrors clips with probability 0.5. Models are evaluated
241 after the fixed training period, without early stopping.

242 **A.3 Test prediction and feature agreement**

243 For prediction, we compare each trained encoder with the same encoder at its random starting
244 weights. Both encoders remain fixed while separate regressions are fitted to their features. Left/right
245 features are averaged over the same valid times, and all regressions use the same feature width and
246 tuning procedure. Four inner groups of training videos select the regression penalty without using
247 test videos. These groups tune only the regression; their inputs can already have been used to train
248 the encoder.

249 The sign-reversal rule uses the difference between features from an original clip and its mirrored
250 copy (Section 3.4). The regression has no intercept, and scaling preserves zero without subtracting a
251 feature mean. Together, these choices ensure that mirroring produces an equal-magnitude prediction
252 with the opposite sign, for either trained or starting-weight features. We also examined two readouts
253 not carried into the main comparison: an odd-part readout that keeps its intercept, which relaxes the
254 exact sign guarantee, and an even-part readout on Φ^+ , which discards the sign-reversing information.

255 Neither changed the conclusions, and both are omitted from Table 1 to keep the reported contrasts
 256 to the direct and sign-fixed readouts.

257 Feature agreement is measured directly, without a prediction model. We match left/right landmarks
 258 between the original and mirrored clips, then divide their squared feature difference by their com-
 259 bined squared magnitude:

$$q(X) = \frac{\|Z(MX) - SZ(X)\|_C^2}{\|Z(MX)\|_C^2 + \|SZ(X)\|_C^2}.$$

260 $Z(X)$ contains the encoder features across all 33 landmarks, and MX is the mirrored clip. S
 261 exchanges left/right landmark positions without changing feature channels; C selects positions valid
 262 in both clips after landmark matching. The score requires at least eight matched positions and a
 263 denominator that is not near zero. No extra transformation is fitted to improve agreement. The score
 264 has no units; zero means matching features, and larger values mean greater disagreement. Features
 265 that change little between clips can also agree closely, so the score must be considered alongside
 266 prediction accuracy. It has no clinical calibration.

267 A.4 Checks and uncertainty

268 Every video receives equal total weight when fitting regressions and calculating scores. For each
 269 initialization, we combine predictions from all five held-out groups before calculating R^2 , then
 270 average the five scores. Disagreement is calculated per clip before averaging with equal video weight
 271 and then across initializations.

272 To estimate uncertainty, we resample the 93 videos 2,000 times, keeping each video’s clips and
 273 initializations together and using the same draw for both conditions. The 95% intervals apply to
 274 each comparison separately and hold the fitted models fixed; they do not include variation from
 275 retraining or different video splits.

276 We checked that test videos stayed out of fitting and tuning, paired encoders shared their starting
 277 weights, and scores gave each video equal weight. Recalculating the scores reproduced the reported
 278 values without rerunning the models. The evaluation plan was finalized after development and was
 279 not externally preregistered.

280 A.5 Reading Figure 1

281 Figure 1 shows the effects summarized in Section 4; this section explains how to read the two panels
 282 and adds one point the main text does not. Panel B is the feature-disagreement comparison and panel
 283 A the prediction comparison, both as differences with pointwise 95% intervals.

284 One feature of panel B is worth drawing out. The reduction from mirrored clips is measured between
 285 the two training variants, yet both variants sit above their untrained references, so those reference
 286 comparisons fall below zero. Mirroring therefore helps relative to training without it, while neither
 287 trained encoder agrees with its mirror as closely as its own random starting weights do. Panel
 288 A carries the prediction story of Table 1: the first two contrasts straddle zero, and the sign-fixed
 289 contrast is the one that turns clearly negative once the prediction rule is fixed the same way for both
 290 encoders.

A. Laterality prediction

Effect of encoder training

Reference: encoder at its random starting weights

Training excludes mirrored clips

Sign reversal not enforced

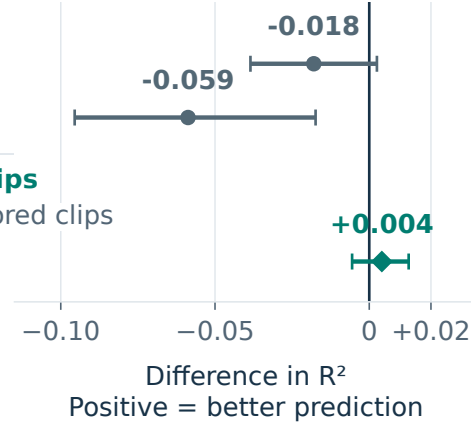
Sign reversal enforced

for both encoders

Effect of adding mirrored training clips

Reference: encoder trained without mirrored clips

Sign reversal not enforced



B. Feature agreement between original and mirrored clips

The encoder's features for the original and mirrored clip are compared after matching left/right body landmarks.

Features compared directly; no prediction rule.

Effect of encoder training

Reference: encoder at its random starting weights

Training without mirrored clips

Training with mirrored clips

Effect of adding mirrored training clips

Reference: encoder trained without mirrored clips

Both encoders trained

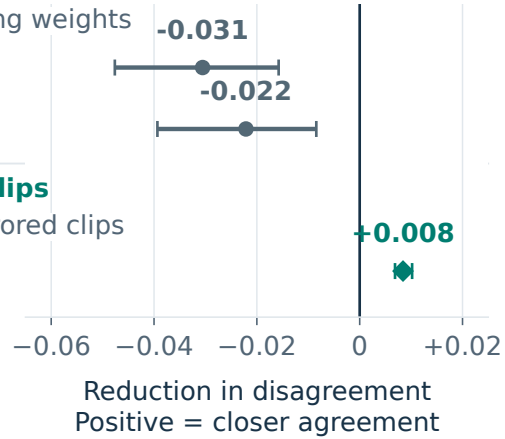


Figure 1: Effects of encoder training and mirrored training clips. Encoder training refers to pretraining the model that converts pose into features. In each panel, the first group compares the trained encoder with the same encoder at its random starting weights. In A, a regression is fitted separately to each encoder's features while encoder weights remain fixed. The first two rows in A exclude mirrored training clips and differ in whether sign reversal is enforced for both encoders. Enforcing it requires equal-magnitude, opposite-sign predictions when left and right are exchanged. The second group in each panel compares encoders trained with versus without mirrored clips. A shows prediction R^2 minus reference R^2 ; B shows reference disagreement minus the tested encoder's disagreement. B compares features directly after matching left/right landmarks, without a prediction rule. Positive values favor the tested condition; zero means no difference. Points average video-balanced paired estimates over five initializations, each covering five held-out video groups. Bars show pointwise 95% intervals from resampling whole videos while keeping fitted models fixed. Intervals crossing zero include possible gains and losses.

Appendix B. Predicting gait-condition labels

This exploratory comparison asks whether learned pose features help distinguish GAVD’s five annotated gait categories. It uses 639 quality-screened clips from 97 YouTube videos, divided into 59 training, 18 validation, and 20 test videos. All clips from a video stay together, with assignments fixed before pose-quality checks. The recordings overlap those in the laterality evaluation and do not provide an independent replication.

The encoder is trained to predict hidden pose features while discouraging nearly constant or redundant features. Training begins with normal-annotated clips, then adds the Parkinson’s, stroke, myopathic, and cerebral-palsy categories in that order, retaining the earlier categories. Annotations determine this schedule, but are not prediction targets during encoder training. The benefit of the ordering was not tested.

We compare summaries of landmark positions and frame-to-frame movement with summaries of the encoder’s features. A third input records how often landmarks are available, without their coordinates. This checks whether missing or poorly detected landmarks carry information about the annotation labels.

A logistic-regression classifier is fitted separately for each input type, with feature scaling and greater weight for less common categories. Encoder weights stay fixed during classifier fitting. Validation videos guide encoder selection and classifier tuning; the final classifiers use all 77 training and validation videos. Test videos remain excluded from these steps.

Table 2: Prediction of dataset annotations on 20 held-out videos. Balanced accuracy is the fraction classified correctly within each category, averaged equally across the five categories. Higher values indicate better performance; scores are rounded to two decimals.

Features used	Videos correctly classified	Balanced accuracy
Pose and movement summaries	10/20	0.44
Learned pose features	6/20	0.26
Landmark availability only	6/20	0.25

Simple pose summaries perform best here, although all three methods misclassify the three stroke-annotated test videos. This supports retaining pose summaries as a baseline. The evidence comes from one split and one training initialization, with no comparison against the same encoder at its random starting weights, so the difference cannot be attributed specifically to pretraining.

Fitting and testing also combine clips differently: fitting uses average features per video, while testing averages the class probabilities predicted for individual clips. This difference can affect the results, as can low-confidence landmark estimates entering the encoder. The exact encoder-training settings have not been independently verified.

318 Appendix C. From video to laterality evaluation

319 A fixed MediaPipe model processes GAVD’s annotated walker crops, producing image-normalized
 320 horizontal and vertical coordinates, estimated relative depth, and visibility scores. These estimates
 321 are not calibrated motion-capture measurements. Laterality calculation and encoder-input prepara-
 322 tion follow separate paths (Appendix A.1).

323 Our S-JEPA adaptation [1] selects valid patches uniformly within the 12 specified landmarks and
 324 hides their content while retaining their positions. The target encoder follows a running average
 325 of the trainable encoder’s weights without receiving gradient updates. For evaluation, we freeze
 326 this target encoder and a copy with its saved random starting weights. Each supplies features to a
 327 separately fitted laterality regression; the pretraining predictor is not used for this step. Additional
 328 regularization uses two unmasked, perturbed views through the trainable encoder (Appendix A.2).

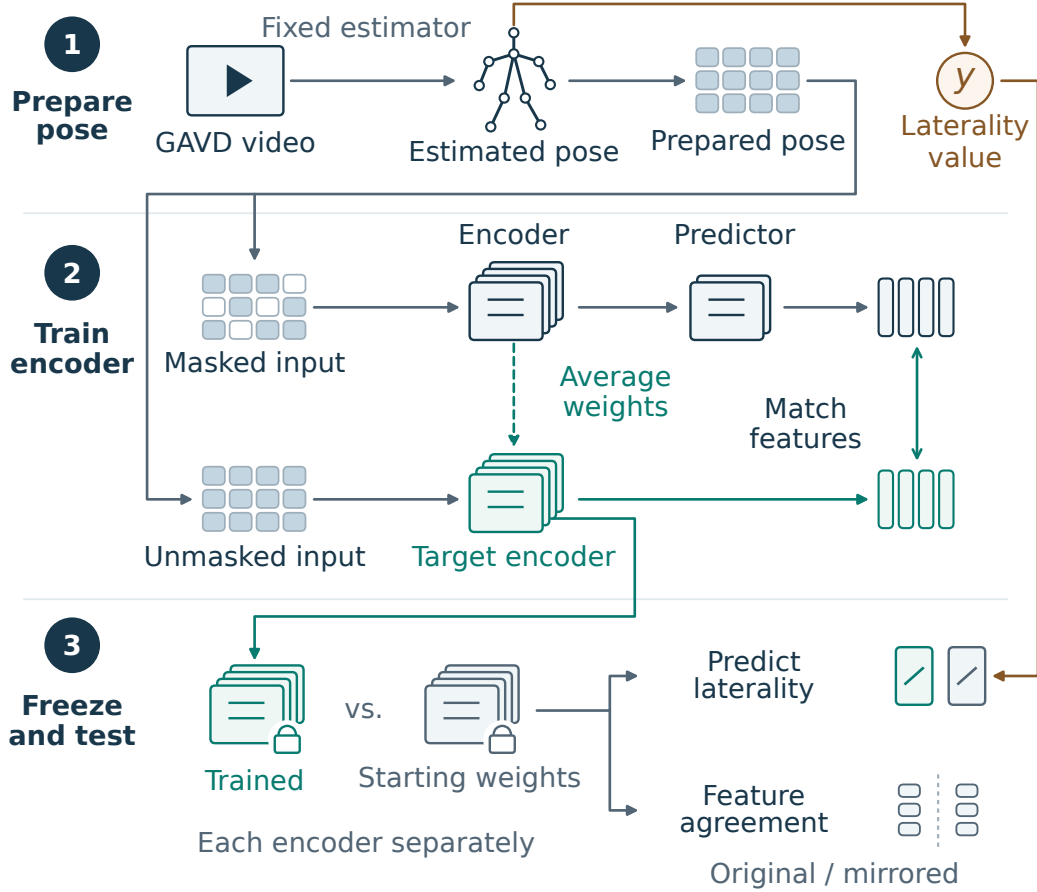


Figure 2: Workflow for training with and without mirrored clips. The gold path carries the laterality value to prediction fitting and scoring, outside encoder pretraining. Each frozen encoder’s regression is fitted separately on training videos. Held-out videos test laterality prediction and feature agreement between original and mirrored clips, with left/right landmarks matched for the feature comparison.

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